1 In triangle ABC, angle BAC is $\frac{\pi}{6}$ radians, angle ABC is $\frac{\pi}{3}+x$ radians. Given that x is sufficiently small, show using the sine rule that

$$\frac{BC}{AC} \approx \frac{1}{\sqrt{3}\left(1-\frac{x^2}{2}\right)+x} \approx a+bx+cx^2 \ ,$$

where a, b and c are constants to be determined. **[4]**

2 A sequence of real numbers u_1, u_2, u_3,… satisfies the recurrence relation

$$u_{n+1} = \left(\frac{n+2}{n}\right)u_n \ , \qquad n \geq 1.$$

(i) Show that $u_k = \frac{k(k+1)}{2}u_1$. **[1]**

(ii) Given that $u_1 = 2$, use the method of mathematical induction to show that

$$S_n = \frac{n}{3}(n+1)(n+2)$$

for all positive integers n, where S_n denotes the sum of the first n terms of the sequence $\{u_n\}$.

[5]

3 **(i)** Solve the inequality $x^2 - \frac{2}{x} \geq \frac{3}{2}x, \quad x \in \mathbb{R}, \quad x \neq 0$. **[2]**

(ii) Without the use of graphic calculator, find the exact value of a such that

$$\int_1^a \left| x^2 - \frac{2}{x} - \frac{3}{2}x \right| \mathrm{d}x = \frac{a^3}{3} - \frac{3a^2}{4} + \frac{1}{4} \quad \textit{where } a \in \mathbb{R},\ a > 2$$ **[4]**

4 The complex number z satisfies the relations $|iz+3| \leq 3$ and $\arg\left(z+\left(3+\frac{3}{i}\right)\right) \geq \frac{\pi}{4}$.

(i) Illustrate both of these relations on a single Argand diagram. **[3]**

(ii) Hence find in exact values, the range of possible values of

(a) $|z-3-3i|$

(b) $\arg(z-3-3i)$ **[3]**

5 Let $\mathrm{f}(x) = \tan^{-1} x$

(i) Sketch the graph of $y = \mathrm{f}(x)$ for $-3 \le x \le 3$. **[1]**

(ii) Find the series expansion of $\mathrm{f}(x)$ in ascending powers of x, up to and including the term in x^3. **[3]**

Denote the answer to **(ii)** by $\mathrm{g}(x)$.

(iii) By substituting $x = \frac{1}{\sqrt{3}}$ into $\mathrm{g}(x)$, show that $\pi \approx \frac{16\sqrt{3}}{9}$. **[1]**

(iv) By substituting $x = \sqrt{3}$ into $\mathrm{g}(x)$, find another approximation for π. **[1]**

(v) Explain why the value of x used for approximation in **(iii)** is better than that in **(iv)**. **[1]**

6 A curve has parametric equations

$$x = at^2,\ y = at^3 \qquad \text{where } t \in \mathbb{R} \text{ and } a > 0.$$

Find, in terms of a,

(i) the equation of the tangent to the curve at the point $\left(\frac{25}{4}a, -\frac{125}{8}a\right)$. **[3]**

(ii) the coordinates of the point where this tangent meets the curve again. **[2]**

(iii) the exact coordinates of the point(s) on the curve at which the normal to the curve passes through the point $\left(\frac{21}{2}a, 0\right)$. **[3]**

7 The functions f, g and $\mathrm{f}^{-1}\mathrm{h}$ are given by

$$\mathrm{f} : x \mapsto \frac{2x^2 + ax - 3}{x+1}, \qquad x \in \mathbb{R}, x > -1 \text{ where } a \text{ is a constant, } a \ne -1,$$

$$\mathrm{g} : x \mapsto (x-1)^2 - 0.25, \qquad x \in \mathbb{R}, 0 < x < 3.5,$$

$$\mathrm{f}^{-1}\mathrm{h} : x \mapsto e^x, \qquad x \in \mathbb{R}, x > 0.$$

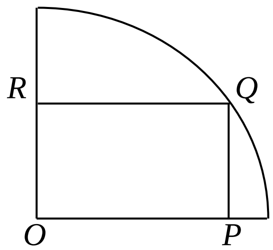
Find the range of values of a for which f has stationary points. Hence, find the set of values of a for which f^{-1} exists. **[4]**

Given that $a = -3$,

(i) find the exact range of fg. **[2]**

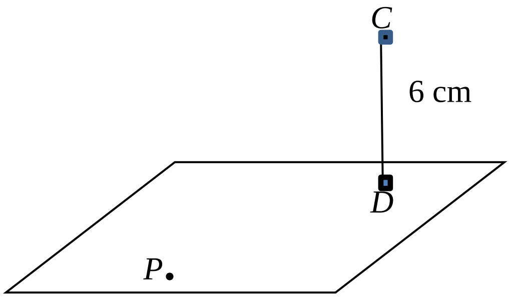
(ii) find h in a similar form. **[2]**

8 (a)



The diagram above shows a rectangle *OPQR* inscribed on the quadrant of a circle of fixed radius *a* with *O* as the center of the circle as shown. If *OP* = *x*, find the area *A* of the rectangle *OPQR* in terms of *a* and *x*. Hence, show that *A* is maximum when the perimeter of the rectangle *OPQR* is 4 times the length of *OP*. **[5]**

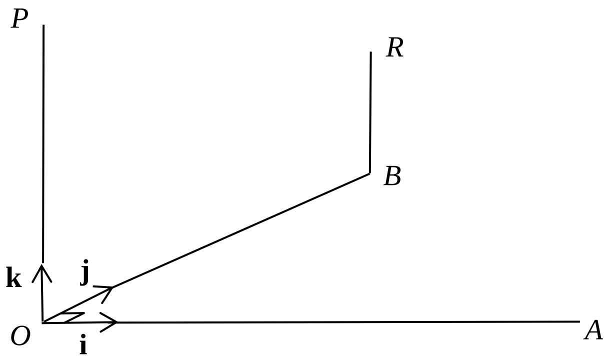
(b)



The line *CD* is perpendicular to a horizontal plane through the point D and *CD* is 6 cm. A variable point *P* moves along a straight line through *D* on the horizontal plane. Given that *P* is moving away from *D* at a speed of 2 cms^{-1}, find the rate of change of the angle *CPD* with respect to time when the distance of *P* from *D* is $6\sqrt{3}$ cm.

[5]

9 In the diagram below, *OAB* is the horizontal base, where *OA* = 6 units, *OB* = 4 units, and $\angle AOB = 90°$. Poles *OP* and *BR* are placed vertically, where *OP* = 5 units and $BR = 2$ units. The unit vectors **i, j, k** are parallel to *OA*, *OB* and *OP* respectively. The point *O* is taken as the origin.



(i) Show that the equation of line *PR* is $\mathbf{r} = \begin{pmatrix} 0 \\ 0 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 4 \\ -3 \end{pmatrix}$, where $\lambda \in \mathbb{R}$. **[2]**

(ii) A point Q divides AB such that $AQ{:}QB = 1{:}3$. Find $\overrightarrow{OQ}$. **[2]**

(iii) A line l with equation $x = 0;\ \dfrac{1-y}{3} = z - 8$ intersects the line PR at a point X. Find the position vector of X. **[3]**

(iv) Let $\mathbf{m}$ be a unit vector along AB. Find $|\ \overrightarrow{QX} \times \mathbf{m}\ |$. Give a geometrical meaning of $|\ \overrightarrow{QX} \times \mathbf{m}\ |$. Hence, or otherwise, find the area of triangle AXB. **[4]**

10 A plane Π_1 is given by the equation $-x + 2y = -5$. A sphere with centre represented by the position vector $-4\mathbf{i} + 3\mathbf{j} + \mathbf{k}$ rests on the plane such that the plane touches the sphere only at one point A.

(i) Find the position vector of A. Hence, or otherwise, find the radius of the sphere exactly. **[4]**

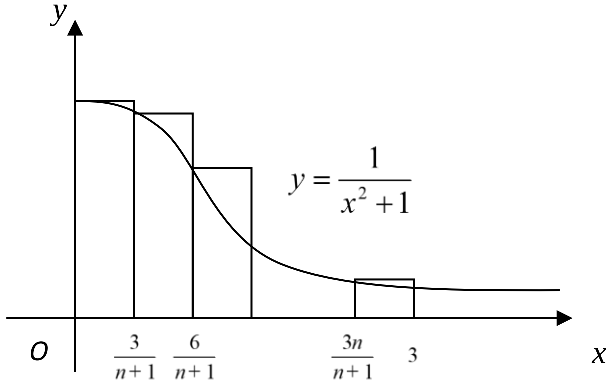
(ii) A line l_1 that passes through the point $a\mathbf{k}$ (where $a > 0$) and the centre of the sphere makes an angle of 30^0 with the plane Π_1. Find a in exact form. **[4]**

(iii) Another plane Π_2 contains the line $\boldsymbol{r} = \begin{pmatrix}4\\3\\1\end{pmatrix} + t\begin{pmatrix}1\\2\\2\end{pmatrix}$, $t \in \mathbb{R}$ and is also parallel to vector $-2\mathbf{i} + 2\mathbf{j} - \mathbf{k}$. Find the vector equation of Π_2 in scalar product form. Hence, show that Π_1 and Π_2 are perpendicular. **[3]**

11 (a) **(i)** The region R is bounded by the curves $y = \dfrac{4}{x^2+1}$, $y = \ln(x+1)$ and the line $x = 1$ and the y-axis. Find the area of region R. **[2]**

(ii) Find the exact volume of the solid formed when R is rotated 2π radians about the y-axis. **[4]**

(b) The diagram shows part of the graph of $y = \dfrac{1}{x^2+1}$.



(i) By considering $n+1$ rectangles of equal width from $x=0$ to $x=3$, show that for all non-negative integers n,

$$A < \sum_{r=0}^{n} \frac{3(n+1)}{9r^2+(n+1)^2},$$

where A is the area bounded by the curve, the axes and the line $x = 3$. **[3]**

(ii) Deduce $\lim\limits_{n\to\infty} \sum\limits_{r=0}^{n} \dfrac{3(n+1)}{9r^2+(n+1)^2}$ exactly. **[2]**

12 (a) By means of the substitution $u=\sqrt{x+1}$, find $\displaystyle\int \frac{2x}{\sqrt{x+1}}\,\mathrm{d}x$. **[4]**

(b) Find $\displaystyle\int \frac{\mathrm{d}x}{(1+x^2)\tan^{-1}x}$ where $x > 0$. **[2]**

(c) **(i)** Differentiate $e^{\sqrt{1-x^2}}$ with respect to x for $|x|$£ 1. **[2]**

(ii) Hence, find the exact value of $\displaystyle\int_0^1 x\,e^{\sqrt{1-x^2}}\,\mathrm{d}x$. **[4]**

End of Paper